Hagyeong Lee

hagyeonglee@postech.ac.kr
+82 010-2497-7351
Website | Scholar | Github | Linkedin

RESEARCH INTEREST

Understanding & designing ML/DL algorithms to handle data with high dimensionality:

Specialized in (1) data compression for efficient transmission, (2) multi-modal vision applications, (3) generative task.

EDUCATION

Pohang University of Science and Technology, POSTECH — Sep. 2022 – Aug. 2024
Master of Science., Major: Electrical Engineering (Advisor: Prof. Jaeho Lee) — Pohang, Korea

- Thesis: Neural Image Compression with Text-guided Encoding for both Pixel-level and Perceptual Fidelity

Ewha Womans University — Mar. 2018 – Aug. 2022
Bachelor of Science., Major: Computer Science Engineering — Seoul, Korea

Incheon Science High School — Mar. 2015 – Feb. 2018
Incheon, Korea

PUBLICATION

(*Equal contribution are denoted by "*".*)

Conference

ICML is the top conference and highly competitive with acceptance rates between 20-30%.

[C2] **Hagyeong Lee***, Minkyu Kim*, Jun-Hyuk Kim, Seungeon Kim, Dokwan Oh, Jaeho Lee, "Neural Image Compression with Text-guided Encoding for both Pixel-level and Perceptual Fidelity," , ***ICML***, 2024.

[C1] **Hagyeong Lee**, Jaeho Lee, "An Empirical Study on the Bias of Generative Image Compression," , *35th Workshop on Image Processing and Image Understanding*, 2023. [Poster]

Journal

[J1] Sujin Yang*, **Hagyeong Lee***, Byounghan Jang, Kee-Deog Kim, Jaeyeon Kim, Hwiyoung Kim, Wonse Park, "Development and validation of a visually explainable deep learning model for classification of C-shaped canals of the mandibular second molars in periapical and panoramic dental radiographs," , ***Journal of Endodontics (IF=4.17)***, 2022. (Featured in a dental newsletter, "DailyDental")

EXPERIENCE

Undergraduate Researcher — Jan. 2022 – Feb. 2022
Efficient Learning Labs (Advisor: Prof. Jaeho Lee) — POSTECH, Korea

- Implicit Neural Representations for Image Compression.

Undergraduate Researcher — Sep. 2021 – Dec. 2021
Center for Spintronics (Advisor: Ph.D. SeokMin Hong) — Post-Silicon Semiconductor Institute, KIST, Korea

- Using machine learning techniques to analyze and predict semiconductor data.

Undergraduate Researcher Sep. 2020 – Feb. 2021
Center for Clinical Imaging Data Science(CCIDS) (Advisor: Prof. HwiYoung Kim) Yonsei Severance Hospital, Korea

- Design an ML model for recognition and classification of dental clinical data

Undergraduate Researcher Jul. 2020 – Aug. 2020
Computer Graphics Lab (Advisor: Prof. YoungJun Kim) Ewha Womans University, Korea

- A Simulation-Based Convergence Content: AI Collaborated with Human-Robot Drawing Model

ML Researcher & Engineer Jan. 2020 – May. 2020
ToonSquare Corp. Seoul, Korea

- Modeling ML System for webtoon creation ; *Text-To-Toon AI*

PATENT

APPARATUS AND METHOD FOR IMAGE ENCODING, AND APPARATUS FOR IMAGE DECODING — KR10-2024-0009324 - Filed

METHOD AND APPARATUS FOR RESTORING IMAGE — KR10-2024-0020643 - Filed

INVITED TALK

Dare To Be Google Developers Groups, Korea, Apr. 2023

- Presented about "From Bias to Balance: Tackling biased datasets towards Generative Models". slide, youtube

Intro. of Neural Image Compression Google Developer Student Club Ewha, Korea, Nov. 2022

- Presented about "Introduction of neural image compression and research career". slide

SCHOLARSHIPS & AWARDS

Full Scholarship Sep. 2018 – Aug. 2022
Awarded for future potential, good grades, and strong academic plans. 미래인재육성재단

- Awarded to only one student in the College of Engineering

Ewha SW Capstone Project, PT 1st , Poster 3rd prize Dec. 2021

2021 Korea Tourism Data Utilization Contest, 3rd prize Nov. 2021
Korean food explanation and recommendation application using image recognition model (OCR) 한국관광공사

Challenge Semester Scholarship Sep. 2020
Deep Learning-based Signal Pattern Design of Abandoned Dog Sound Ewha Womans University.

- Raised and donated funds to help abandoned dogs.

SW Future Scholarship Dec. 2019
Awarded for future potential in SW and CS. Ewha Womans University

Google Developer Student Club Lead Aug. 2019 – Aug. 2021
Organize and manage programs (Tech Seminar, DevFest, Hackathon, Pep talk) Google Korea

Young Engineers Honor Society Jul. 2019
245th Forum National Academy of Engineering of Korea (한국공학한림원)

PROJECTS

Visual signal compression algorithm using large language models. June. 2023 – May. 2024

- Designed Text-Adaptive Compression, a new compression system that utilizes textual information with visual signals (images) to achieve high accuracy and perceptual quality.
- funded by the Samsung Advanced Institute of Technology (SAIT), Korea.

AI interpolation for low resolution with Lissajous scanning in cancer histology Sep. 2023 – Feb. 2024

- Developing interpolation model to achieve 2 µm resolution based on 10Hz high-speed Lissajous scanning micro-probes.
- collaborated by VPIX, Korea
- funded by the Ministry of SMEs and Startups (MSS), Korea.

TEACHING EXPERIENCES

[EECE695D-01] Efficient Machine Learning Systems Mar. 2024 – June. 2024

Working as a teaching assistant for Prof. Jaeho Lee.

[EECE236] Learning about Elec. Eng. Using Matlab Mar. 2023 – June. 2023

Working as a teaching assistant for Prof. Sunggu Lee.

BACKGROUND

Language: English (Advanced)

Programming Language: Python (Advanced), C, C++, JavaScript (Intermediate)

Library: PyTorch, OpenCV

Workflow: Git, Docker, Notion

REFERENCE

Prof. Jaeho Lee, Professor, POSTECH
Relationship: M.S. advisor
E-mail: jaeho.lee@postech.ac.kr

Ph.D Jun-Hyuk Kim, Researcher, SAIT
Relationship: Collaborator during text-adaptive compression project.
E-mail: jh131.kim@samsung.com

Ph.D SeokMin Hong, Researcher, KIST
Relationship: Collaborator during a research internship at a post-silicon semiconductor institute.
E-mail: shong@kist.re.kr

Prof. Hwiyoung Kim, Professor, Yonsei Severance Hospital
Relationship: Undergraduate advisor during a research internship at CCIDS.
E-mail: hykim82@yuhs.ac